

Dashboard Demonstration Script

Customer Segmentation Dashboard — 5-10 Minute Live Demo

Demonstration Flow

This script follows the 10-step demonstration flow: Open Application → Executive Dashboard → Apply Filters → Analyze Segments → Compare Two Segments → Search Individual Customer → Generate/View Segment → Show Marketing Recommendation → Demonstrate Invalid Input → Explain Business Value.

1. Open Application

"We'll start by launching the dashboard with a single command: `streamlit run app/app.py`. This opens in the browser automatically — no separate installation or configuration needed beyond the one-time pip install."

2. Executive Overview

"First, we review the Executive Overview. This is the landing page and gives leadership the headline numbers at a glance: 2,237 total customers, 4 segments, \$606 average spend, and a 14.9% campaign response rate — everything a manager needs in the first five seconds."

What to point out

The Customer Segment Distribution donut and Revenue Contribution bar chart side-by-side — this immediately shows the revenue-concentration story: a small segment drives most of the revenue.

3. Apply Customer Filters

"Now let's filter down to just customers aged 40-60 with income above \$60,000, using the sidebar. Watch how every chart and table on the page updates immediately — there's no 'Apply' button to click, it's fully live."

What to point out

The filter-match counter at the bottom of the sidebar, which shows exactly how many customers match the current selection.

4. Analyze Customer Segments

"Moving to section 7, Customer Segments, we can see the size, average income, spend, and recency for every segment in one table, plus an expandable characteristics panel for each."

5. Compare Two Segments

"Now the Segment Comparison section. Let's compare High-Value Customers against Discount Seekers. The radar chart makes the contrast immediately visible — High-Value leads on every dimension except family size. We can also download this comparison as a CSV for a follow-up meeting."

6. Search Individual Customer

"Let's look up a specific customer. I'll enter Customer ID 5524 and click Look Up Customer. Immediately we see their segment — High-Value Customers — along with their income, spend, age, and a tailored marketing recommendation."

7. Generate/View Segment (New Customer)

"Now the more powerful feature: let's analyze a brand-new customer who isn't in our database yet. I'll enter Age 35, Income \$50,000, Total Spending \$1,200, 5 web purchases, 8 store purchases, and Recency 20 days, then click Analyze Customer. The dashboard runs this through our actual trained K-Means model live — not a lookup table — and assigns a segment in real time."

8. Show Marketing Recommendation

"Notice the output includes not just the segment, but the Customer Value, Retention Risk, and a specific Recommended Action, Channel, and product focus — this is the Module 08 business logic wired directly into the live prediction."

9. Demonstrate Invalid Input

"Let's test what happens with bad input. I'll change the age field to -10 and click Analyze Customer again. Instead of crashing, the application shows a clear message: 'Age must be between 0 and 120.' The same happens if I type text like 'abc' into a numeric field — a clear error, no crash."

10. Explain Business Value

"To summarize the business value: this dashboard turns a one-time clustering analysis into a living tool. Marketing can filter and target segments today, look up any customer's profile and recommended action in seconds, and even classify brand-new leads the moment they're captured — all without touching a Jupyter notebook or asking a data scientist."

Closing

Total demonstration time: approximately 7-8 minutes. Leave 2-3 minutes for questions, particularly around the K=4 business-override decision (Chapter 6 of the Final Project Report) and the future-improvements roadmap (Chapter 12).